STRESSES IN CYLINDERS AND COMPOSITE TUBES (THIN-WALLED PRESSURE VESSELS)





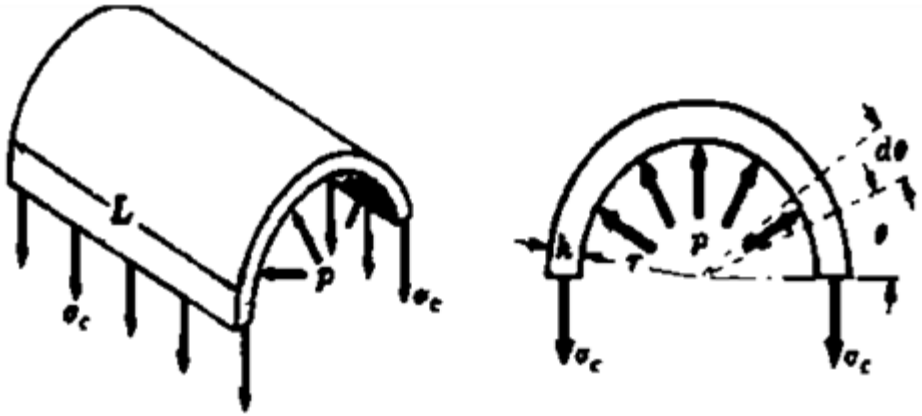




THIN-WALLED PRESSURE VESSELS

The internal pressure acting on the long sides of the cylinder will give rise to a circumferential stress in the wall of the cylinder this is known as hoop stress. If the ends of the vessel are closed the pressure acting on the ends is transmitted to the walls of the thin cylinder producing a longitudinal stress. Here, radial stresses can be neglected. We are to consider thin walled vessels. That is cylinders with the ratio of thickness to internal diameter less than 1:20.

Hoop Stresses



Let us consider a section through a diameter of cylinder of internal radius r and length L.

Resolving forces vertically on the part of the material upon which the pressure *p* is acting, we have

$$F_v = pA = p.2r.L$$

Similarly, Resolving forces vertically on the part of the material *r*esisting this force, we have; $F_v = 2Lh\sigma_h$

$$= 2prL \quad \therefore \sigma_h = \frac{pr}{h}$$

This implies

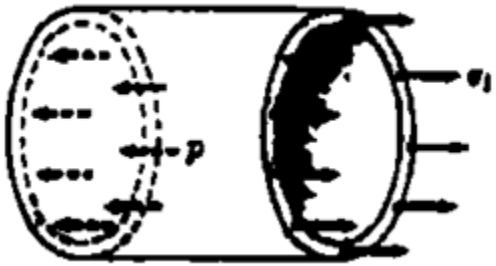
$$2Lh\sigma_h = 2prL \quad \therefore \sigma_h = \frac{pr}{h}$$

Longitudinal Stresses

Now consider any transverse cross - section of the cylinder far removed from the ends. When the ends are closed, the longitudinal force on each closed end is giving as

$$\sigma_L = p.A = p.\pi r^2$$

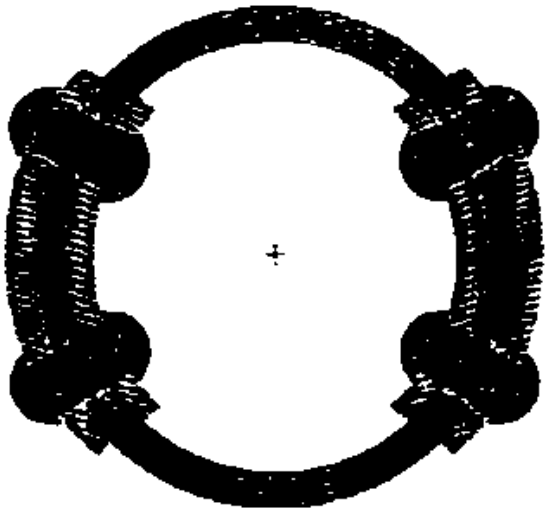
and the transverse cross - section is resisted by the longitudinal stress as follows



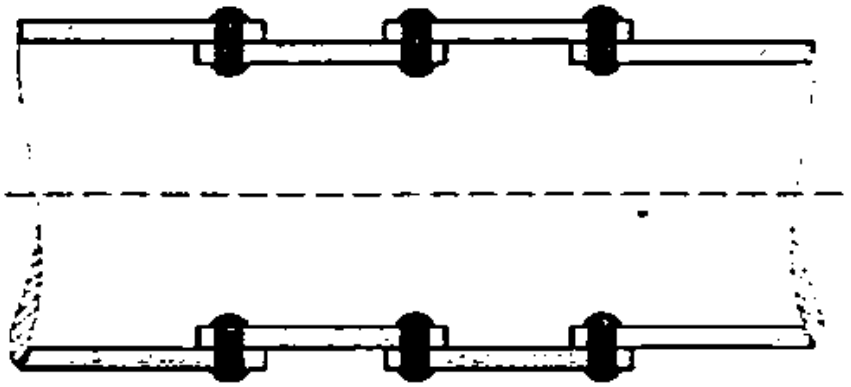
$$\sigma_L.2\pi rh = p\pi r^2 \quad \therefore \sigma_L = \frac{pr}{2h} = \frac{\sigma_h}{2}$$

NOTE. If η is the efficiency of the riveted joints of the shell(ie $\eta \neq 100\%$), then stress $\sigma_L = \frac{pr}{2\eta h} = \frac{\sigma_h}{2}$

TYPES OF JOINTS



(a) Joining by Butt Joint



(b) Joining by Lap Joint

Example 1

A thin wall cylinder has an internal diameter of 20 cm and is 0.5 cm thick. It is subjected to an internal pressure of 3.5 MN/m^2. Estimate the hoop and longitudinal stress of the ends if the cylinder ends are closed

Solution

$$\sigma_L = \frac{pr}{2h} = \frac{(3.5)(10)}{2(0.5)} = 35MPa$$

$$\sigma_h = \frac{pr}{h} = \frac{(3.5)(10)}{0.5} = 70MPa$$

Strains in the Walls of the vessels

The resulting strains, hoop and longitudinal strains are as follows

$$\varepsilon_h = \frac{1}{E}\left(\sigma_h - \upsilon\sigma_L\right) = \frac{pr}{Eh}\left(1 - \frac{1}{2}\upsilon\right)$$

$$\varepsilon_L = \frac{1}{E}\left(\sigma_L - \upsilon\sigma_h\right) = \frac{pr}{Eh}\left(\frac{1}{2} - \upsilon\right)$$

The circumference of the cylinder has increased by a small amount $2\pi r\varepsilon_h$

The increase radius is therefore $r\varepsilon_h$

The increase in length of a unit length of the cylinder is ε_L

The change (increase) in internal volume of a unit length of the cylinder is $\delta V = \pi\left(r + r\varepsilon_h\right)^2\left(1 + \varepsilon_L\right) - \pi r^2.1$

$$\frac{\delta V}{\pi r^2} = \left(1 + \varepsilon_h\right)^2\left(1 + \varepsilon_L\right) - 1$$

Strains in the Walls of the vessels

since $\varepsilon_h^2, \varepsilon_L^2, \varepsilon_{\mathrm{h}}\varepsilon_{\mathrm{L}}$ are negligible

$$(1+\varepsilon_h)^2(1+\varepsilon_L)-1$$

$$=(1+2\varepsilon_h)(1+\varepsilon_L)-1=2\varepsilon_h+\varepsilon_L$$

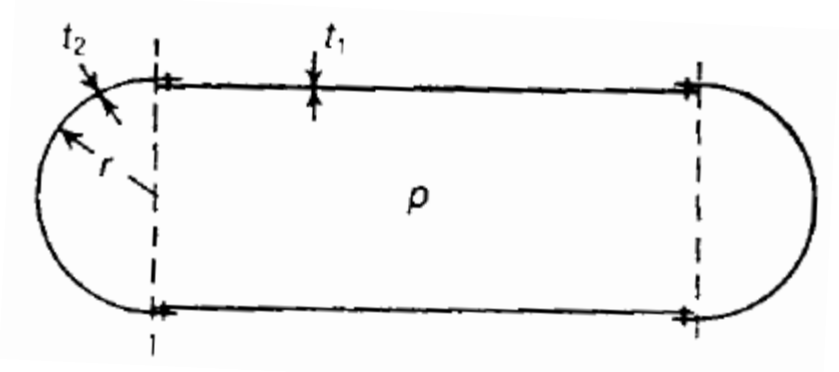
$$2\varepsilon_h+\varepsilon_L=\frac{pr}{Eh}\left(\frac{5}{2}-2\upsilon\right)$$

Hence the volumetric strain is

$$\frac{\delta V}{\pi r^2}=\frac{pr}{Eh}\left(\frac{5}{2}-2\upsilon\right)$$

To find the increase in volume it is necessary to multiply the volumetric strain by the original volume.

Hemispherical Ends on a Cylindrical Shell



For the cylindrical shell portion,

we have; Hoop stress and strain :

$$\sigma_h = \frac{pr}{t_1} \text{ and } \varepsilon_h = \frac{pr}{Et_1}\left(1-\frac{1}{2}\upsilon\right)$$

Longitudinal stress and strain :

$$\sigma_L = \frac{pr}{2t_1} \text{ and } \frac{pr}{Et_1}\left(\frac{1}{2}-\upsilon\right)$$

For the Hemispherical section we have

$$\sigma_h = \sigma_L = \frac{pr}{2t_2}; \text{due to symmetry}$$

The Hoop strain is $\varepsilon_h = \frac{pr}{2Et_2}(1-\upsilon)$

for no distortion of the junction under pressure. $\varepsilon_h = \varepsilon_L$;

Thus, $\frac{pr}{Et_1}\left(1-\frac{1}{2}\upsilon\right) = \frac{pr}{2Et_2}(1-\upsilon)$

$$\frac{1}{t_1}\left(1-\frac{1}{2}\upsilon\right) = \frac{1}{2t_2}(1-\upsilon) \Rightarrow \frac{t_2}{t_1} = \frac{1-\upsilon}{2-\upsilon}$$

If the Poisson ratio is given as 0.3, then

$$\frac{t_2}{t_1} = \frac{7}{17}$$

Bulk Modulus K

If a hydrostatic pressure *p* (is equal in all direction) acting on a body of initial volume V, causes a reduction in volume equal to δV, then the bulk modulus K is defined as the ratio between the fluid pressure and volumetric strain.

$$K = \frac{-p}{\delta V / V}$$

The negative sign indicate reduction in volume.

Example 2

A boiler drum consists of a cylindrical portion 2m long, 1m diameter, and 25mm thick is closed by a hemispherical ends. In a hydrostatic test to 10 N/mm^2, how much additional water will be pumped in, after initial filling at atmospheric pressure? Assume the circumferential strain at the junction of cylinder hemisphere is the same for the drum material. E=207,000 N/mm^2; ν=0.3, for water K=2100 N/mm^2

Solution

For cylinder, the hoop stress is

$$\sigma_h = \frac{pr}{t} = \frac{(10)(500)}{25} = 200\, N/mm^2$$

Longitudinal stress is

$$\sigma_L = \frac{pr}{2t} = \frac{(10)(500)}{2(25)} = 100\, N/mm^2$$

Hoop strain is

$$\varepsilon_h = \frac{1}{E}(\sigma_h - \upsilon\sigma_L) = \frac{1}{E}(200 - 0.3(100)) = \frac{170}{E}$$

$$\varepsilon_L = \frac{1}{E}(\sigma_L - \upsilon\sigma_h) = \frac{1}{E}(100 - 0.3(200)) = \frac{40}{E}$$

Solution

Increase in volume for the cylindrical portion :

$$\Delta V_1 = (2\varepsilon_h + \varepsilon_L)V = \left[2\left(\frac{170}{E}\right) + \frac{40}{E}\right]V = 380\frac{V}{E}$$

$$V = \frac{\pi d^2 L}{4} = \frac{\pi}{4}(1m)^2(2m) = 1.57x10^9\, mm^3$$

$$\Delta V_1 = 380\frac{V}{E} = 380\left(\frac{1{,}570{,}000{,}000}{207{,}000}\right) = 2.88x10^6\, mm^3$$

For the two hemispherical ends,

$$\Delta V_2 = (2\varepsilon + \varepsilon)V = \left[2\left(\frac{170}{E}\right) + \frac{170}{E}\right]V = 510\frac{V}{E}$$

$$\varepsilon = \varepsilon_h$$

$$V = \frac{4\pi d^3}{24} = \frac{\pi}{6}(1m)^3 = 0.5x10^9\, mm^3$$

Solution

Decrease in volume of water due to pressure

$$\Delta V_3 = \frac{p}{K} V = \frac{10}{2100}\left[\frac{(\pi)(1000)^2(2000)}{4} + \frac{\pi(1000)^3}{6}\right]$$

$$= 10x10^6\, mm^3$$

Total additional water required

$$\Delta V_T = \Delta V_1 + \Delta V_2 + \Delta V_3 = 14.2x10^6\, mm^3$$

Strain in Pressure Vessels

The stresses in the vessels are hoop stress σ_h and *longitudinal* stress σ_L. The hoop strain is

$$\varepsilon_h = \frac{1}{E}\left(\sigma_h - \nu\sigma_L\right)$$

$$\sigma_h = \frac{pr}{t};\quad \sigma_L = \frac{pr}{2t}$$

Increase in the circumferential length

$$2\pi r\varepsilon_h$$

$$2\pi r\varepsilon_h = \frac{2\pi pr^2}{Et}\left(1-\frac{1}{2}\upsilon\right)$$

Total *circumferential* length

$$2\pi r + \frac{2\pi pr^2}{Et}\left(1-\frac{1}{2}\upsilon\right)$$

We divide by

$$r + \frac{pr^2}{Et}\left(1-\frac{1}{2}\upsilon\right)$$

$$\text{Increase in the radius} = \frac{pr^2}{Et}\left(1-\frac{1}{2}\upsilon\right)$$

In the case of a very long pipe or an open ended pipe where the longitudinal stress, σ_L is equal to zero, the increase in radius will be $\frac{pr^2}{Et}$

Problem 1

Consider two laminated pressure vessel with an initial interference on diameter of 0.25 mm. The thickness of the inner shell is 2.5 mm, and that of the outer shell is 2.0 mm. The outer cylinder is heated, placed on the inner and allowed to cool, thus providing a ‘shrink fit’. If both cylinders after cooling assume a common diameter of 100 mm find the interface pressure if both shell are made of steel. E=207,000 N/mm^2

SOLUTION

Increase in the radius of the outer shell is

$$I_o = \frac{pr^2}{Et} = \frac{p(50)^2}{2E}$$

Decrease in the inner shell is

$$I_i = \frac{pr^2}{Et} = \frac{p(50)^2}{2.5E}$$

Interference on radius

$$I = I_o + I_i = \frac{p(50)^2}{2E} + \frac{p(50)^2}{2.5E} = 0.125$$

Problem 2

Consider a thin steel cylinder junction fitting over copper inner cylinder as shown. Find the hoop stresses in each shell due to a temperature rise of 30 K. Assume there is no longitudinal stress

Solution

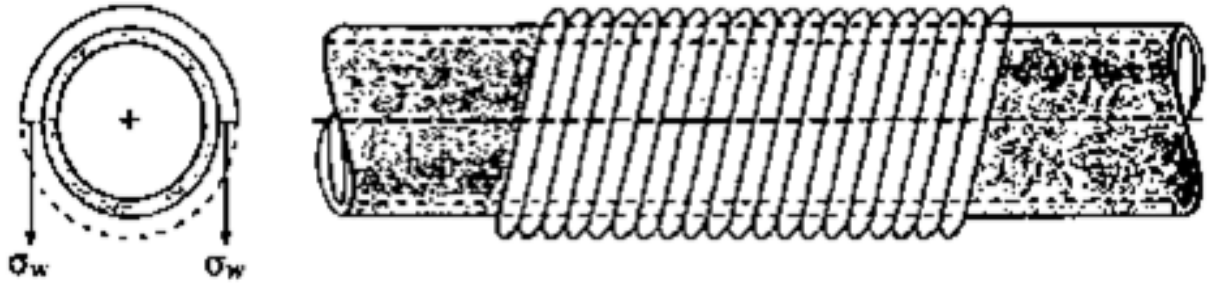
First consider the two cylinders are heated separately and then find the interference (Net) on radius. Then assume a pressure p that will reduce the radius of the inner shell and increase the interference diameter of the outer shell so that they will have a common radius therefore the interference will vanish

Wire-Bound Thin Cylindrical Shells

- Sometimes, we have to strengthen a cylindrical shell against bursting in longitudinal section *(i.e.,* due to hoop or circumferential stress).

- This is done by winding a wire under tension, closely round the shell as shown in Fig. 31.6.

- Its effect will be to put the cylinder wall under initial compressive stress.



Example 6-9:

- A *cast iron pipe of 300 mm internal diameter and 12 mm. thick is wound closely with a single layer of circular steel wire of* 5 *mm diameter under a tension of 60 MPa. Find the initial compressive stress in the pipe section. Also find the stresses set up in the pipe and steel wire when water under a pressure of* 4 *MPa is admitted into the pipe. Take E for cast iron and steel as 100 GPa and 200 GPa respectively. Poisson's ratio = 0.3.*

Solution

- Given: Diameter of pipe (d) = 300 mm; Pipe Thickness (t) = 1.2 mm; Diameter of wire = 5 mm; Tension in wire = 60 MPa = 60 N/mm^2; Pressure of water (p) = 4 MPa = 4 N/mm^2 Modulus of elasticity for cast iron (E_c) = 100 GPa = 100 x 10^3 N/mm^2; Modulus of elasticity for steel (E_s) = 200 GPa = 200 x 10^3 N/mm^2; and Poisson's ratio (1/m) = 0.3

Initial compressive stress in the pipe section

- A little consideration will show that the pipe will be subjected to compression (due to tension in the wire) before the water under pressure is admitted into the pipe. Now consider one mm length of the pipe. We know that number of wire sections for 1 mm pipe length

$$n_w = \frac{2}{Diameter\ of\ \ wire} = \frac{2}{5} = 0.4$$

- The initial compressive force in the wire *(i.e.,* before the water under pressure is admitted into the pipe),

$$F_{cw} = n_w x A_w x \sigma_w = 0.4 x \left[\frac{\pi}{4} x(5)^2 \right] x60 = 471.3\ N$$

- Therefore, initial compressive stress in the section,

$$\sigma_c = \frac{F_{cw}}{2t} = \frac{471.3}{2x12} = 19.6\ N/mm^2$$

ASSIGNMENT

1. A thin spherical copper shell of diameter 0.3 m and thickness 1.6 mm is just full of water at atmospheric pressure. Find how much the internal pressure will be increased by pumping in 25,000 mm^3 of water.
   E=100,000 N/mm^2; $\nu=0.286$; $\mathrm{K}=2200\ \mathrm{N}/mm^2$.

2. A long steel tube, 7.5 cm internal diameter and 0.15 cm thick, is plugged at the ends and subjected to internal fluid pressure such that the maximum direct stress in the tube is 120 MN/m^2. Assuming v = 0.3 and E = 200 GN/m^2, find the percentage increase in the capacity of the tube.

ASSIGNMENT

3. A bronze sleeve of 200 mm internal diameter and 6.4 mm thick is pressed over a steel liner 200 mm external diameter and 16 mm thick, with a force-fit allowance of 0.07 mm on diameter. Treating both as thin cylinders find (1) the radial pressure at the common surface, (2) the hoop stresses, (3) the percentages of the fit allowance met by the sleeve and liner.